

PERSONAL INFORMATION

Name: Jingkai LAN

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EDUCATION BACKGROUND

University of Bern (Master) — SWITZERLAND

September 2023 - September 2025

Major: Bioinformatics and Computational Biology

Overall Grade: 5.0 / 6.0

Shihezi University (Bachelor) — CHINA

September 2019 - June 2023

Degree: Bachelor of Science in Biological Science

GPA: 3.64 / 4.0

SKILL

Language Proficiency: English; Chinese**Computer skills:** Python, R, Bash, HPC, Deep learning, Machine Learning, NGS analysis, Data analysis.**Experiment skills:** Nucleic Acid Extraction, Gel electrophoresis, PCR, Spectrophotometer, Clean bench.

RESEARCH PROJECTS

Advancing metal-binding protein predictions with deep learning

February 2025 - September 2025

Thomas Lemmin Lab, IBMM, University of Bern — Master's thesis

- Metalloprotein annotations remain incomplete in major biological databases (e.g., PDB, UniProt).
- The goal of this project is to use deep learning to predict potential metal-binding proteins solely from sequence inputs to support annotation, thereby reducing the cost of experimental and manual annotation.
- Integrated existing metalloprotein resources and introduced an innovative strategy using in silico mutagenesis at metal-binding sites (ProteinMPNN) to generate non-metal-binding proteins, with stringent cleaning and quality control using metal-binding site information and structure prediction tools (e.g., ESM, CHAI1).
- Trained multiple models to predict metal binding and enzyme-related functional properties using supervised / semi-supervised deep learning, as well as embedding-based machine learning.
- Using sequence-only inputs, the trained models achieved performance comparable to leading approaches on this task (including methods that leverage structural information or engineered features), without requiring extensive feature engineering.

NeurIPS – Open Polymer Prediction 2025

June 2025 - September 2025

University of Notre Dame — NeurIPS 2025 competition track

- Addressed data limitations in polymer discovery by developing models to predict key polymer properties directly from chemical structures.
- Trained machine learning models using SMILES-only inputs to predict target polymer properties.

- Extracted molecular graph features and fingerprints with RDKit; applied multiple approaches including XGBoost, LightGBM, and ChemBERT, with systematic hyperparameter optimization.
- Achieved a ranking within the top third of the final leaderboard.

Diversity of Peach Tree Viruses in Xinjiang, China

November 2021 - August 2022

Baiming Cui Lab, Shihezi University — **Bachelor's Thesis**

- Conducted bioinformatics analysis of metatranscriptomic data from *Prunus persica* var. *compressa* leaf samples collected in the Shihezi region to identify viral species.
- Profiled viral abundance and distribution patterns across samples.
- Performed phylogenetic analyses to characterize genetic relationships among detected viral strains.
- Identified multiple peach-associated viruses and investigated their genetic diversity and evolutionary status in Xinjiang, providing evidence to support local peach disease management.

Morphological and Molecular Survey of Bat Species in Shihezi, Xinjiang

November 2020 - August 2021

Abduriyim Shamshidin Lab, Shihezi University — **Mini project**

- Collected naturally deceased bats in the Shihezi region (Xinjiang) and conducted morphological identification.
- Extracted genomic DNA from muscle tissue samples.
- Amplified mitochondrial ND1 and Cyt b genes via PCR and performed sequence alignment against the NCBI database.
- Confirmed species identity as *Pipistrellus pipistrellus* and provided the first combined morphological and molecular evidence for its presence in the region.

Publication:

Morphological and molecular confirmation of the common pipistrelle bat, *Pipistrellus pipistrellus* Schreber, 1774 (Vespertilionidae: Chiroptera), in Xinjiang, China.

Abduriyim, S.*, Kasimu, T., **Lan, J.**, Pu, Z., Bai, J. & Wang, Y. (2022).

Mammalia, vol. 86, no. 3, pp. 298–302. DOI: doi.org/10.1515/mammalia-2021-0045

PRIZES AND AWARDS

Excellent Graduate (one of the seventeen award-winners among 180 students)

December 2022

*** Summer Camp Excellent Participant Award, ShanghaiTech University**

July 2022

First-class scholarship in the university (one of the five receivers among 104 students)

December 2021

* ('Summer Camp' means admission program of China university, offering excellent students exemption from postgraduate entrance exams.)